



Prototype of a Wireless Seismic Data Acquisition System

Course EE 411 (Senior Design Project)

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Abstract

Cables are commonly used in conducting seismic surveys. Figure 1 illustrates a typical seismic survey where a vibrator generates mechanical seismic waves (swept from 2Hz to 90Hz) that are refracted and reflected by the different underground sublayers enabling the identification of their material composition (Figure 2). Sensing these waves is done by a grid distributed sensors called Geophones. High costs and heavy loads are introduced when the number of cabled sensors becomes very large as shown in Figure 3.

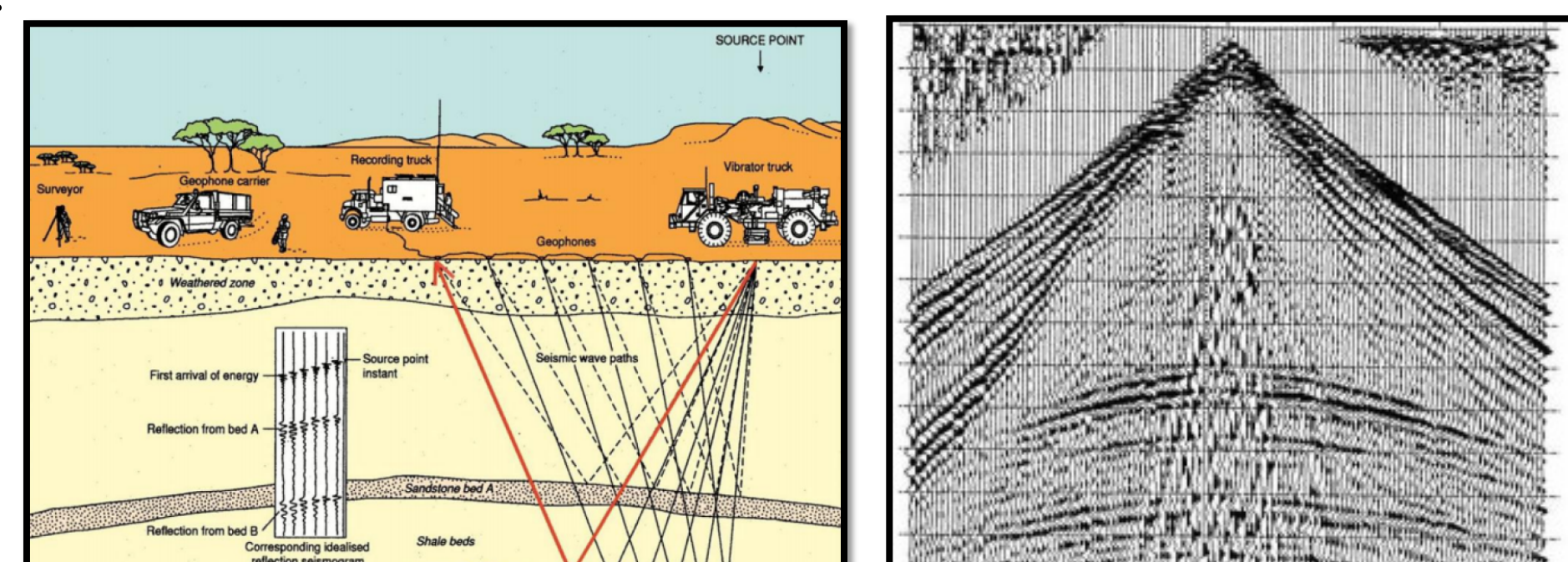


Figure 1. Typical Seismic Survey.

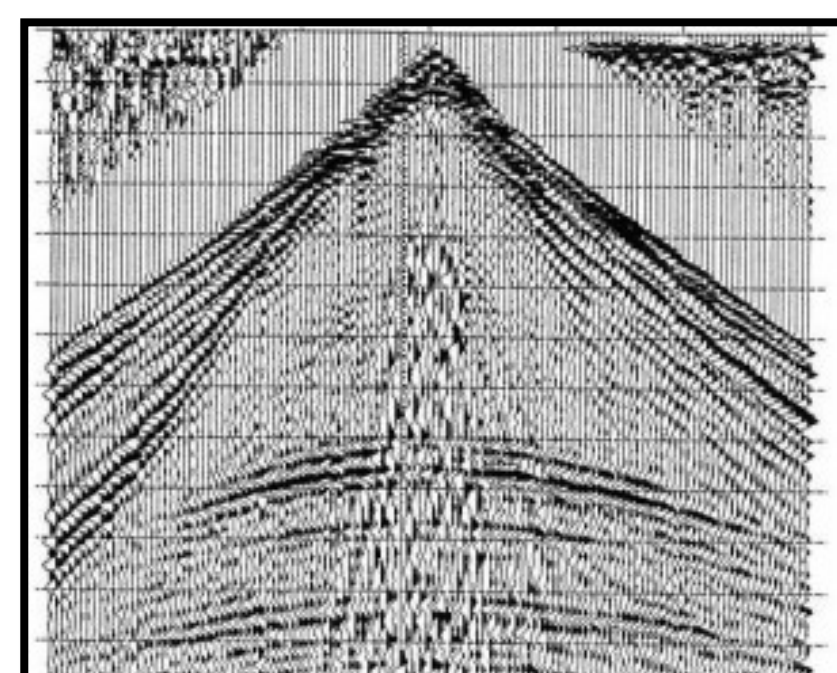


Figure 2. Seismic Data.



Figure 3. Cabled Sensors.

To avoid such drawbacks, researchers propose to do the complete data acquisition process wirelessly. This project implement a prototype of a wireless seismic data acquisition system. Data are received from a coil-based geophone (magnet rounded by a coil to sense the seismic vibrations by Faraday's law of induction). These data pass through conditioning circuits and converted to digital words and finally transmitted through air.

Key Words

Coil-Based Geophone, Signal Conditioning, Analog to Digital Conversion,, Microcontrollers, Wireless Communication.

Objectives

- Offer alternative method to replace wired seismic data acquisition system.
- Examine 3 deferent seismic sensors.
- Implement an applicable signal conditioning circuit for the parameters of seismic signals with low power consumption.
- Test different communication techniques (Serial & wireless).
- Present a wireless seismic data acquisition prototype with emphasized low power consumption and cost.

Methodology

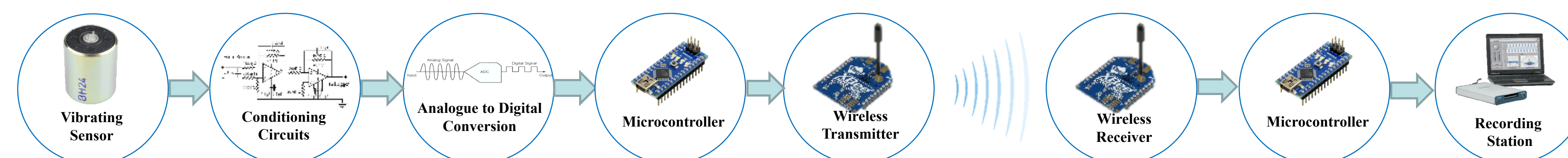


Figure 4. Graphical building blocks of the system.

Sensor

The impulse response of the used Coil-Based Geophone sensor was experimentally obtained by applying a very quick and strong hummer strike is shown in Figure 5.

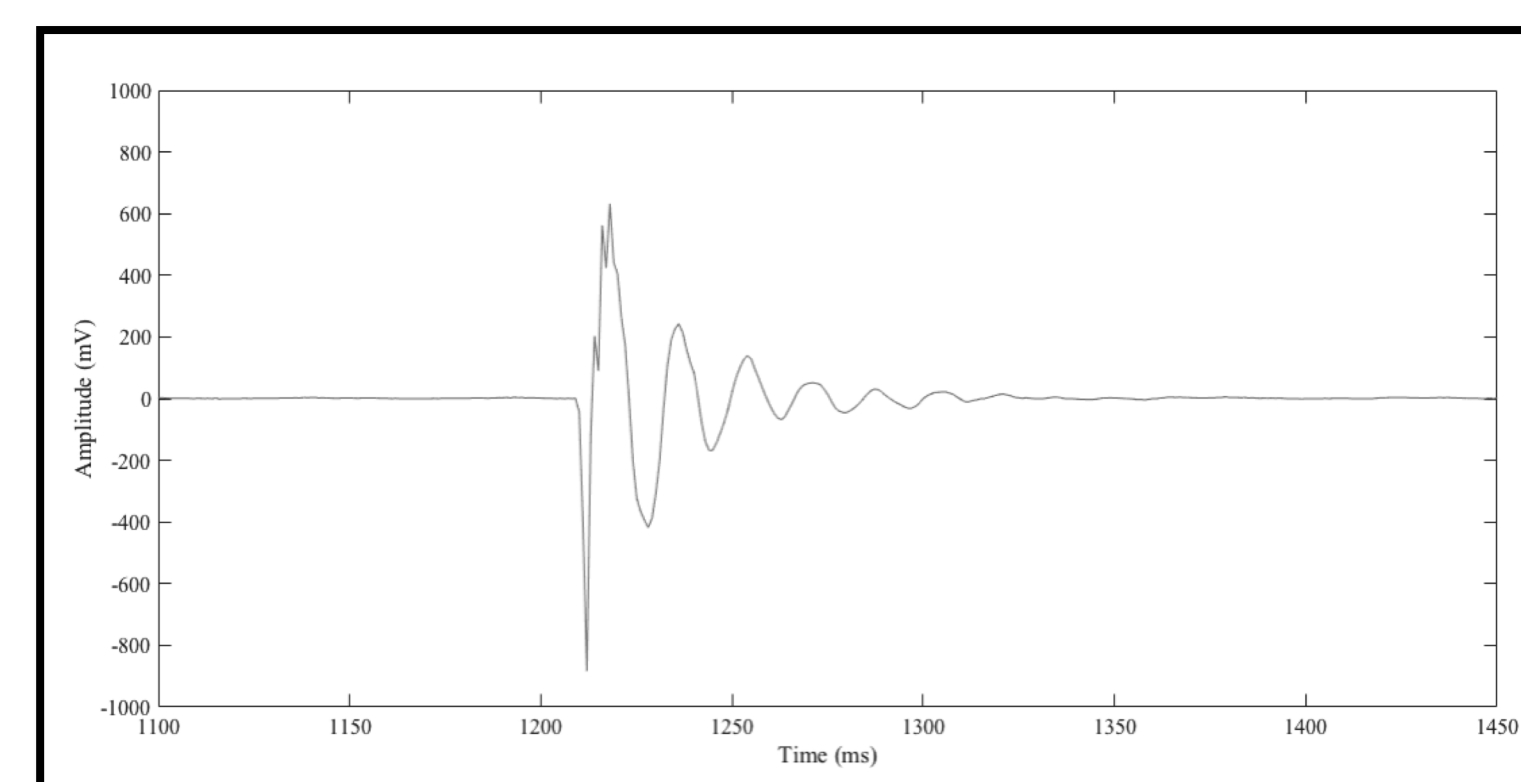
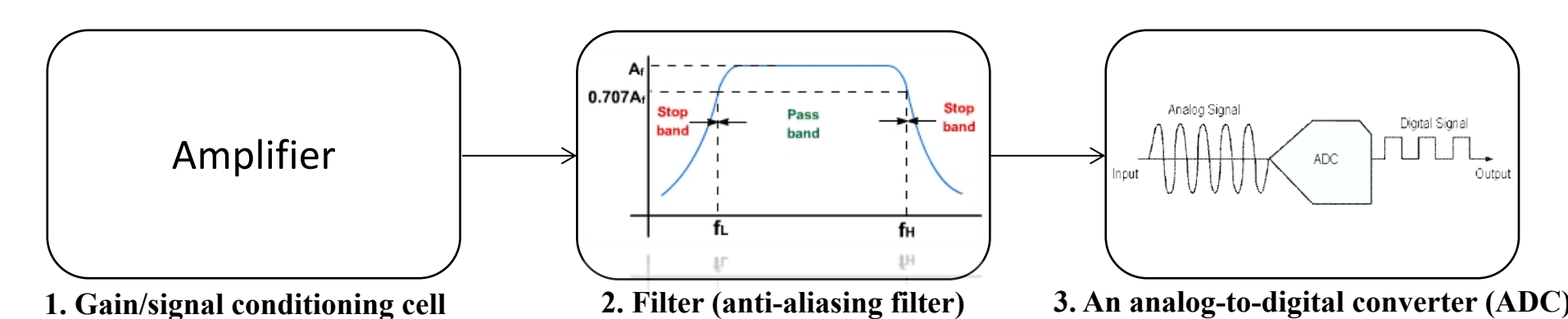


Figure 5. Impulse response of the used sensor.

Signal Conditioning Circuit

Signal conditioning circuit main parts:



Five conditioning stages were implemented to enhance the signal. The stages are shown and labeled in the Figure 6.

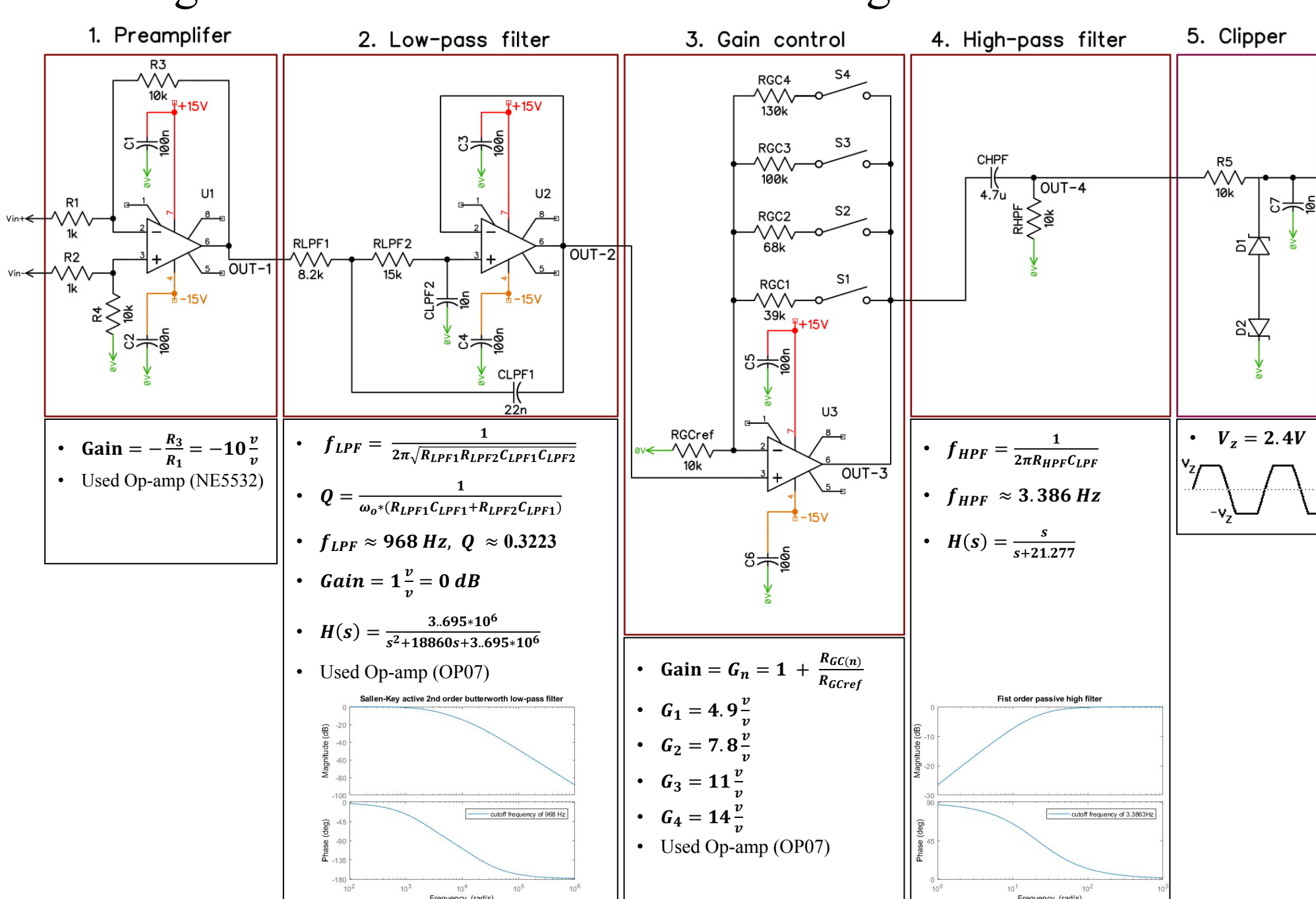


Figure 6. Signal conditioning circuits.

Analog to Digital Conversion

ADC conversion based on delta sigma was implemented as shown in Figure 7. The converter chip was set to do the following:

- 24 bits conversion.
- Sampling rate of 2ms.
- Accept differential input.
- Initial self-calibration.
- Unity gain.

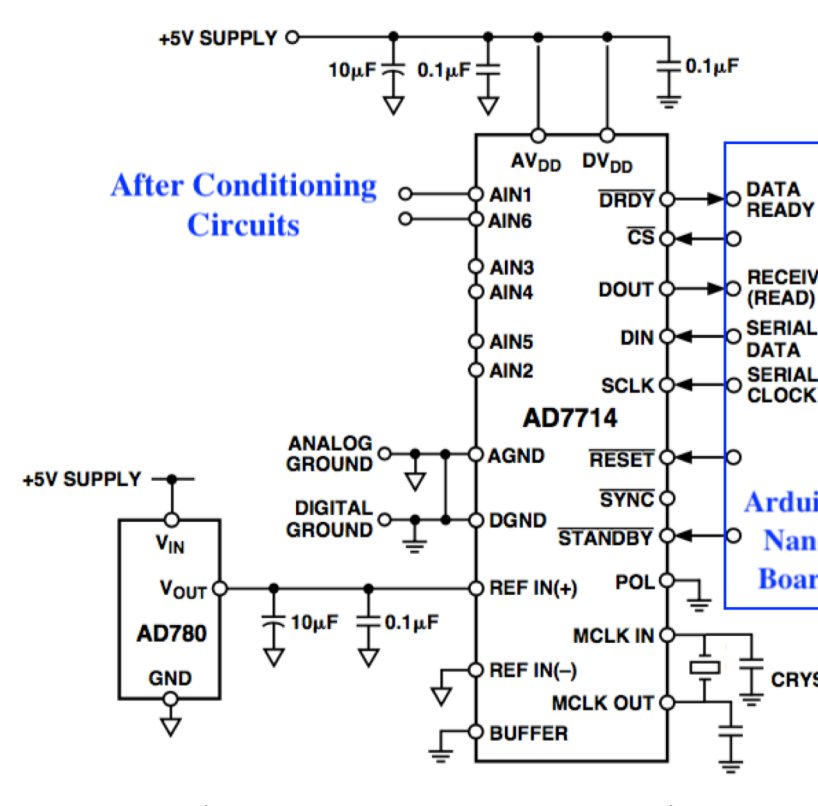


Figure 7. ADC connections.

Microcontroller

Arduino Nano manages the operations of the ADC and the wireless communication. Figure 8 describe the used algorithm to obtain the collected data from the ADC. After configuring the desired settings, the DRDY (data ready) pin is checked repeatedly to assure the end conversion.

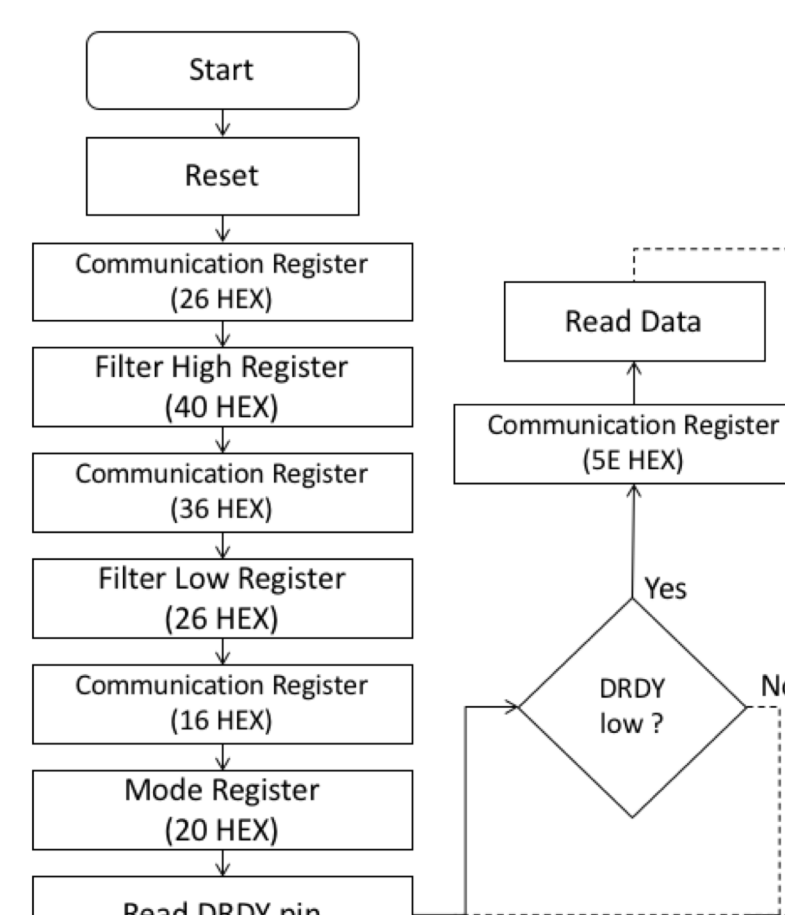


Figure 8. Used algorithm.

Wireless communication

The RF module uses Zigbee standard which is for low-power, low-cost and wireless mesh-networking satisfying the survey requirements. To verify the suitability of the used RF modules, a sweep of sinusoidal signal according to the generated seismic frequencies was sent and received wirelessly. Table 1 summarizes the obtained results. Figure 9 shows two samples in the frequency domain.

Sent Frequency	Received Frequency	Sent Frequency	Received Frequency
2 Hz	2.0 Hz	35 Hz	35.2 Hz
5 Hz	5.0 Hz	50 Hz	50.1 Hz
8 Hz	8.1 Hz	60 Hz	60.2 Hz
10 Hz	10.0 Hz	70 Hz	70.3 Hz
15 Hz	15.1 Hz	80 Hz	80.2 Hz
20 Hz	20.1 Hz	90 Hz	90.3 Hz
25 Hz	25.2 Hz	---	---

Table 1. Wirelessly sent and received frequencies.

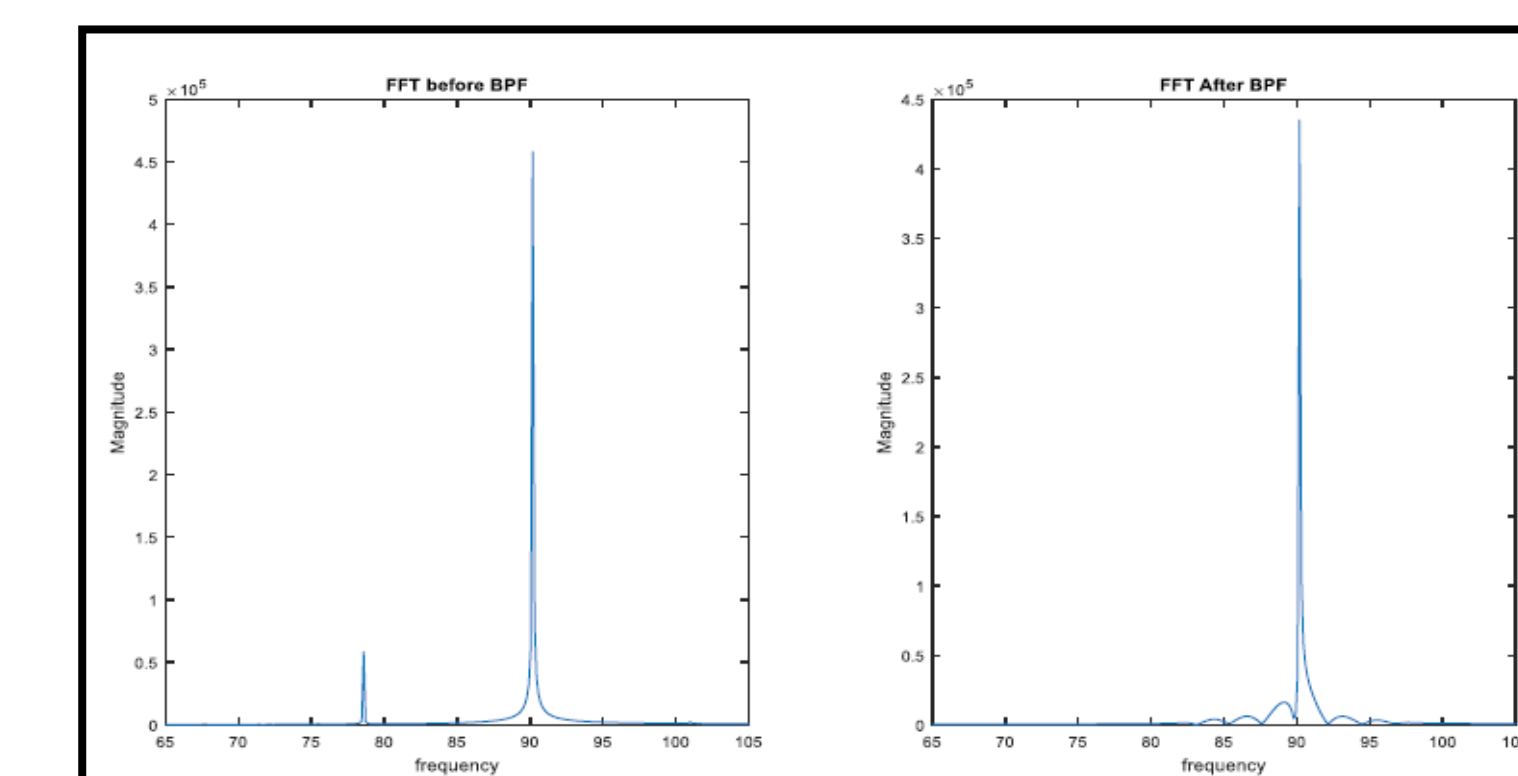


Figure 9: Samples of sent and received signals in frequency domain

Results

To verify the operation of this design, without pre-ADC conditioning circuit, a hummer experiment was conducted in cooperation with geophysics department. A vibrating sensor was connected, in parallel, to both our designed system and their optimized system to compare results. Figure10 shows the experimental setup. Figures11-12 show results.

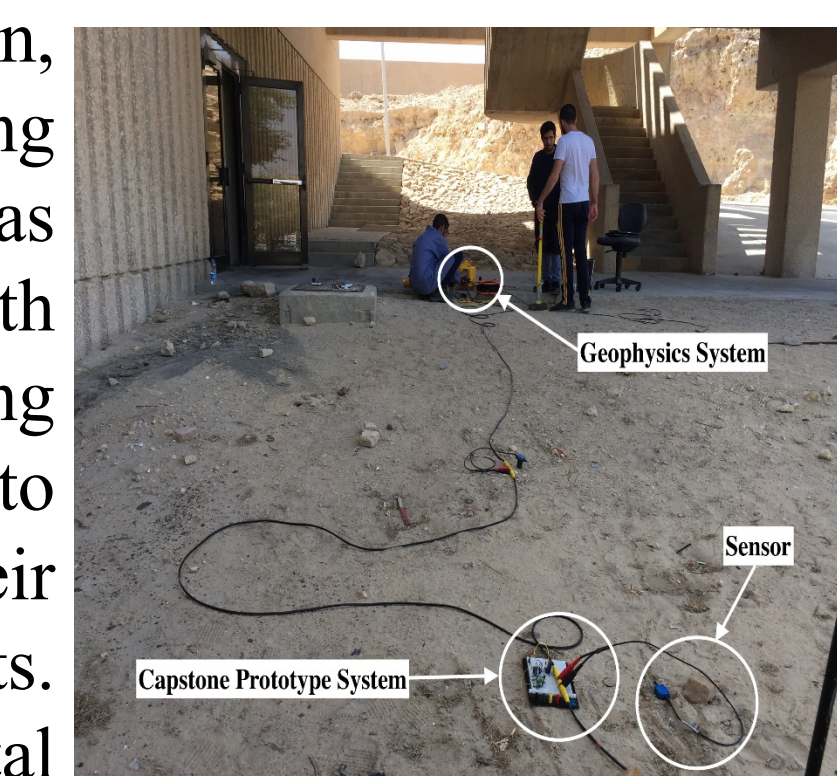


Figure 10: Hummer experiment setup

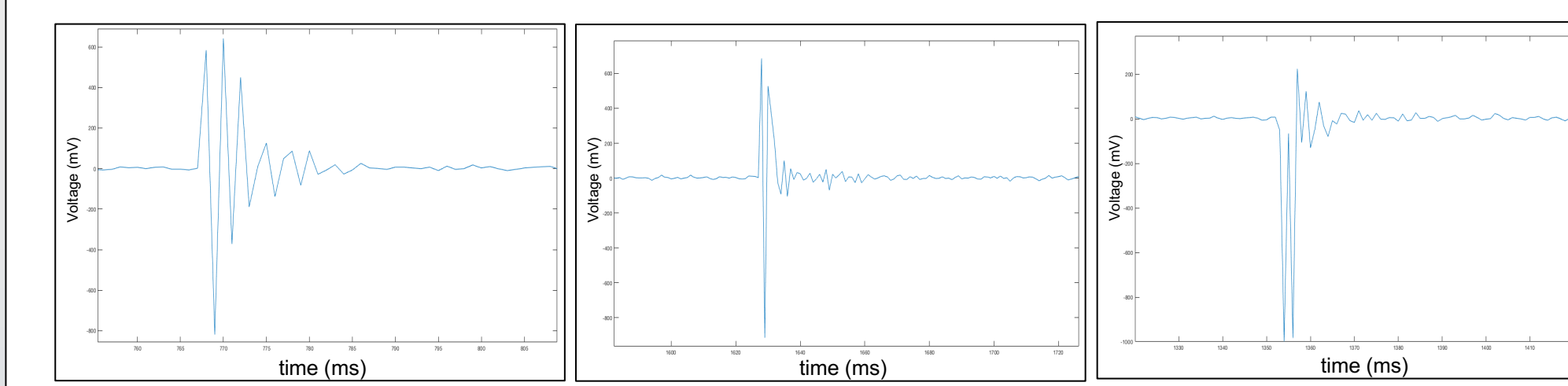


Figure11: Results of wirelessly received signal with our system. (a), (b) and (c) correspond to collected data with 10m, 25m and 50m away from transmitter, respectively

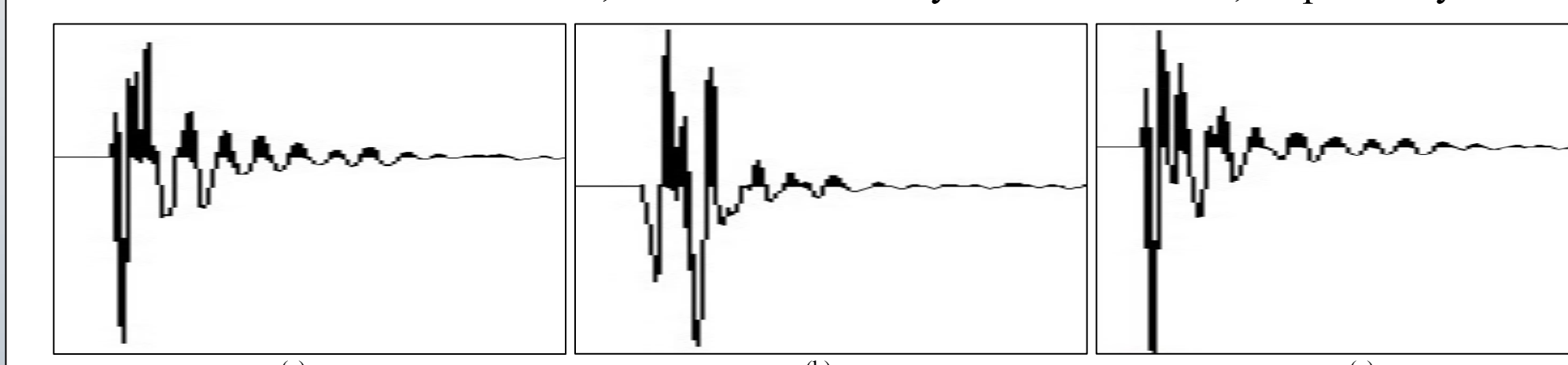


Figure12: Results of the optimized seismic data acquisition system. (a), (b) and (c) correspond to collected data of Figure10 (a), (b) and (c), respectively

Conclusions

- The prototype can be connected to existing geophone products.
- It has comparable results to existing products.
- It costs less than 400 SR.

Recommendations

- Use multiple geophones and nodes.
- Reduce the ADC resolution to lower the power dissipation
- Enhance the microcontroller code and adding frequency reconfigurability to optimize the operations and reduce the costs.

References

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